# Efficient On-Orbit Remote Sensing Imagery Processing via Satellite Edge Computing Resource Scheduling Optimization

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***Abstract*— With the enormous scale of remote sensing imagery generation, on-orbit computing has become a crucial paradigm to enable near-real-time processing. Due to the limited onboard resources and on-orbit power supply, satellite edge computing (SEC) is developed for satellite–ground collaboration, aiding on-orbit computation. However, the intermittent satellite-to-ground transmission link poses an efficiency challenge when collaborating SEC resources. Therefore, this article proposes a satellite edge computing resource scheduling technique for on-orbit remote sensing imagery processing (SECORS). First, we design a remote sensing mission-specific SEC architecture, which involves an offline–online satellite working mode. Subsequently, a computational resource scheduling model (SEC-RSM) is established, including the directed acyclic graph (DAG) model and mathematical problem formulation. Next, to obtain effective scheduling solutions, we develop an end-to-end algorithm leveraging the multiagent proximal policy optimization and heuristic rule of the earliest finish time (SEC-MPH). Finally, we build a simulation SEC platform to carry out experiments and implement several methods as the comparison including multiobjective evolutionary algorithms, deep reinforcement learning approaches, and the scheme without optimization (baseline). Simulation results show that SECORS achieves 68.87% and 66.60% reductions in time and energy for on-orbit computation. Moreover, our method improves the energy efficiency ratio (EER) by three times and achieves high processing capacity with 548 pixels per unit of power (W) and time (ms).**

***Index Terms*— Multiagent deep reinforcement learning (MADRL), on-orbit computation, remote sensing imagery processing, resource scheduling model, satellite edge computing (SEC).**

## NOMENCLATURE

| Notation | Description |
|---|---|
| $\mathcal{S}$ | Set of satellites and satellite server in the constellation. |
| $\mathcal{T}$ | Set of all on-orbit tasks generated from satellites, where $i, j \in \mathcal{T}$. |
| $\mathcal{T}^+$ | Set of total tasks, and $\mathcal{T}^+ = \mathcal{T} \cup \{0, 0'\}$. Here, 0 and $0'$ indicate the entry task and the exit task, respectively, for each device. |
| $\mathcal{G}$ | Set of ground device (cloud server). |
| $\mathcal{K}$ | Set of all devices (computational resources) in SEC, where $k, l \in \mathcal{K} = \mathcal{S} \cup \mathcal{G}$. |
| $\mathcal{P}$ | Set of task pairs. $(i, j) \in \mathcal{P}$ indicates that task $i$ is the predecessor of task $j$ in DAG. |
| $\mathcal{Q}$ | Set of device pairs. $(k, l) \in \mathcal{Q}$ indicates $k \in \mathcal{G}$ and $l \in \mathcal{S}$, or $k \in \mathcal{S}$ and $l \in \mathcal{G}$. |
| $w_i^k$ | Time cost of processing task $i$ on device $k$. |
| $d_{ij}^{kl}$ | Time cost of data transmission from device $k$ (processing task $i$) to device $l$ (processing task $j$). |
| $t_i^{\text{start}}$ | Start time of task $i$. |
| $t_i^{\text{end}}$ | Completion time of task $i$. |
| $t_i^{\text{tran_start}}$ | Transmission start time of task $i$. |
| $t_i^{\text{tran_end}}$ | Transmission completion time of task $i$. |
| $t^{\text{off_start}}$ | Start time of communication interrupt between satellites and ground station. |
| $t^{\text{off_end}}$ | Completion time of communication interrupt between satellites and the ground station. |
| $p_k$ | Running power of device $k$. |
| $p^{\text{tran}}$ | Power of data transmission. |
| $M$ | Sufficiently large positive number. |
| $C^{\text{T}}$ | Total time cost for executing on-orbit tasks. |
| $C^{\text{E}}$ | Total energy cost for executing on-orbit tasks. |
| $x_{ij}^k$ | Decision variables. If tasks $i$ and j are scheduled on device $k$, and $i$ is a predecessor of $j$, 1; otherwise, 0. |
| $y_{ij}^{kl}$ | Decision variables. If tasks $i$ and $j$ are scheduled on devices $k$ and $l$, respectively, 1; otherwise, 0. |
| $z_i$ | Auxiliary variables. |

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## I. INTRODUCTION

REMOTE sensing imagery processing is a crucial technique that provides various satellite services, such as ocean and farm surveillance, and emergency management for disasters [1], [2], [3]. The conventional satellite pattern is transmitting the collected data to the ground and then processing it. However, with the increasing equipment of new high-rate data acquisition detectors on satellites, e.g.,

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ultrawide cameras and hyperspectral sensors, this mode brings about severe delays in the processing of large-volume remote sensing data. Therefore, on-orbit computation is proposed to achieve the real-time or near-real-time performance by reducing transmission overhead. Meanwhile, the deployment of high-performance and low-power embedded chips improves onboard computational power. European Space Agency (ESA) has launched OPS-SAT, a nanosatellite containing more powerful onboard computers for on-orbit applications, and developed the first onboard deep neural network demonstrator for Earth observation $\phi$-sat-1 [4]. National Aeronautics and Space Administration (NASA) has designed SpaceCube edge-node intelligent collaboration (SCENIC) [5] to demonstrate the next-generation artificial intelligence computing capacity with several embedded AI devices, such as tensor processing units (TPUs) and field-programmable gate arrays (FPGAs). According to this paradigm, several research has been conducted on on-orbit remote sensing imagery processing [6], [7], [8].

Our previous work [7] has studied on-orbit imagery processing based on a heterogeneous multiprocessor system and achieved a promising performance. The proposed methodology is based on a single-satellite on-orbit computing mode. By analyzing the experimental results, we found that this approach started to degenerate when the image size exceeded $2048 \times 2048$. Therefore, the single-satellite onboard computational resources are still not enough to process large-scale remote sensing imagery. In light of this, we investigate satellite edge computing (SEC) which incorporates more resources from satellites and ground devices. SEC architecture represents a paradigm shift in the way data are processed in space-based systems, integrating computational resources at the edge of the satellite-terrestrial network. The primary components of SEC include satellites equipped with onboard processing units and terrestrial infrastructure, where intersatellite and satellite-to-ground communication links support interconnection between different devices. SEC is particularly beneficial for satellite applications requiring low-latency remote sensing imagery processing, through multisatellite and satellite–ground collaborations. Achieving effective collaborations in SEC poses a challenging endeavor, thereby necessitating a comprehensive study of the computational resource scheduling problem.

However, two main challenges exist in leveraging SEC resource scheduling to improve the efficiency of on-orbit remote sensing imagery processing. From the perspective of computation, on-orbit remote sensing imagery processing is characterized by compute-intensive patterns. This places a huge computational burden on the resource-limited and energy-constrained satellite, necessitating SEC resource scheduling optimization. Besides, it requires an abstract task description to model on-orbit missions in the SEC resource scheduling issue. Remote sensing imagery processing missions are usually implemented by complex algorithms, such as cloud detection, geometric correction, and image classification, which generate the complicated on-orbit task modeling and further increase the difficulty of the SEC resource scheduling problem.

On the communication side, compared with one satellite, communication costs incurred by intersatellite and satellite-to-ground transmissions need to be considered during SEC resource scheduling. However, the satellite-to-ground transmission link is unstable due to limited connectable periods incurred by orbital characteristics, which also suffers from the constraint caused by ground station deployment. Meanwhile, the high-throughput on-orbit data pose an obstacle for the bandwidth-limited satellite-to-ground link. The acquisition rate of remote sensing data could be up to approximately 200 Gb/s [7], but the current commercially used satellite constellation, as exemplified by Starlink, just provides the maximum downlink transmission rate of 220 Mb/s [9], which is far lower than the data generation rate. Thus, the communication limitations make the SEC resource scheduling problem complex and hard to acquire favorable solutions. Given the above analysis, how to effectively schedule SEC resources for on-orbit remote sensing imagery processing, considering computation, communication, and energy consumption has become an intractable issue worthy of further investigation.

Motivated by efficient on-orbit remote sensing imagery processing, we propose an SEC resource scheduling optimization method, called satellite edge computing resource scheduling for on-orbit remote sensing imagery processing (SECORS). Overall, our method introduces several innovative aspects.

1) We develop an end-to-end SEC resource scheduling scheme based on multiagent deep reinforcement learning (MADRL) to meet the real-time or near-real-time on-orbit processing. This method incurs less overhead compared with traditional metaheuristic optimization techniques.
2) In comparison to existing MADRL approaches, SECORS is designed from a problem-specific SEC resource scheduling scenario considering the satellite offline–online mode and practical remote sensing imagery processing missions.
3) We incorporate a heuristic strategy called earliest finish time (EFT) into the proposed method, which improves the scheduling capacity through a feasible task execution sequence on the individual device.

The contributions of this article are as follows.

1) First, this work devises an on-orbit remote sensing imagery processing-based SEC architecture. To mitigate the intermittent satellite-to-ground link caused by orbital motion patterns, we propose an offline-online satellite working mode in SEC. Three scenarios are presented according to various potential offline periods during satellite task execution.
2) Aiming at efficient on-orbit computation, we establish an SEC resource scheduling model SEC-RSM. Initially, four typical remote sensing missions, imagery slicing, cloud detection, geometric correction, and classification, are chosen as the case study. Next, we use the DAG to construct abstract satellite task description based on the algorithmic analysis. Furthermore, an SEC resource scheduling problem considering the online–offline mode

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is formulated, with the objective of minimizing time and energy costs.

3) Finally, an MADRL algorithm SEC-MPH is developed to obtain effective scheduling solutions. Based on Markov decision process (MDP) analysis of the proposed problem, we leverage the multiagent proximal policy optimization (MAPPO) to implement cooperative training for multiple agents. In addition, a scheduling heuristic called EFT is adopted to determine the execution sequence of multiple tasks assigned to the same device. By building a simulation SEC platform and conducting experiments, we identify that the proposed technique can make on-orbit remote sensing imagery processing more efficient than the scheme without optimization.

The remainder of this article is organized as follows. Section II reviews the related literature. Section III constructs SEC-RSM including satellite task model based on DAG and computational resource scheduling problem formulation. Section IV designs SEC-MPH to address the problem. Section V builds a simulation SEC system and conducts numerical experiments to verify the effectiveness of the proposed methodology. Finally, Section VI concludes this work and presents some potential future research topics.

## II. Related Work

Although edge computing has been widely used in terrestrial-level service scenarios, it is still in infancy to investigate SEC for on-orbit applications, such as remote sensing imagery processing. Therefore, this section analyzes the development of SEC, resource scheduling model, and solution algorithms in the related literature to clarify the research value of this study.

### A. Satellite Edge Computing

In recent years, SEC is developed to cope with the proliferation of satellite data and on-orbit computation requirements. A considerable amount of work [10], [11] has been proposed to explore new application modes based on SEC and to address various challenges. Kong et al. [12] introduce a pioneering secure satellite-edge computing framework specifically designed for low Earth orbit (LEO) satellites. It seeks to solve the intricate problems of confidentiality, authentication, and intellectual property rights in collaborative power grid applications, particularly in the context of real-time line outage detection. Li et al. [13] point out the power problem in SEC and proposes a battery-aware energy optimization framework, which uses a pattern-aware online energy scheduling algorithm to achieve sublinear regret and minimal constraint violations in the dynamic and uncertain wireless environment. In [14], concerning the timeliness of streaming analytics in SEC networks, an optimization method is presented to mitigate the unique challenges posed by dynamically moving satellites and uncertain data volumes. As for SEC computation peer offloading issues, Zhang et al. [15] leverage Lyapunov optimization and distributed decision-making strategy to efficiently balance uneven workloads and minimize latency and energy cost. In addition, several network-based problems are also investigated for the SEC framework, such as function placement [16], and server and service selection [17]. In contrast to the preceding studies, Zhu et al. [18] choose hyperspectral image processing in SEC as the research object and design a novel two-tier collaborative band selection framework using a multiagent reinforcement learning approach. The current work is mainly focused on issues related to the basic SEC architecture of computing and network but rarely dedicated to computational resource scheduling problems.

### B. Resource Scheduling Model in Edge Computing

For generality, two issues need to be tackled when studying edge computing resource scheduling problems: abstract task description and problem formulation. Previous research has summarized that modeling DAG is an explicit method to describe those tasks running as computer programs. In [19], the researchers construct DAG for fundamental operators in secure multiparty computation. Avallone and Ashraf [20] abstract forwarding paradigm tasks using DAG, which is used to address the challenges in software-defined networks. Hu et al. [21] apply DAG to automotive systems where the functionality is modeled. Meanwhile, there are some standard DAGs modeled from general algorithms and stochastic DAGs modeled without any practical implications. They are designed to form generic test instances (e.g., Cybershake, Gaussian elimination, Stencil, and Epigenomics) for algorithm evaluation. Such DAGs are used in [22] to estimate the proposed energy-aware real-time scheduler on heterogeneous computing systems. Similarly, Xu et al. [23] also select these DAG models to validate the effectiveness of a new multitree genetic programming approach for dynamic workflow scheduling in fog computing.

A range of resource scheduling problems related to various edge computing scenarios have been investigated in recent years [24], [25]. Seah et al. [26] study the fairness and latency requirements in sliced 5G multiaccess edge computing systems. They present a dedicated scheduling problem considering both communication and computing resources. In [27], a joint computation resource scheduling and service placement problem is formulated to demonstrate how to elaborately select and deploy services on the edge node and how to reasonably allocate computation resources to placed services, according to the arrivals of computation workloads. Another study [28] investigates the collaborative task computing and on-demand resource scheduling issue in vehicular edge computing (VEC) and establishes a mathematical model with the response latency as an optimization objective. Jiang et al. [29] define a multiobjective energy-efficient task scheduling and computational resource allocation problem for edge devices equipped with heterogeneous processors. To efficiently manage task interruptions and energy dynamics in asynchronous wireless-powered mobile edge computing (MEC) environments, Cang et al. [30] formulate a scheduling problem regarding computing resources.

Although many studies have been conducted to explore scheduling problems in edge computing, few of them seek to

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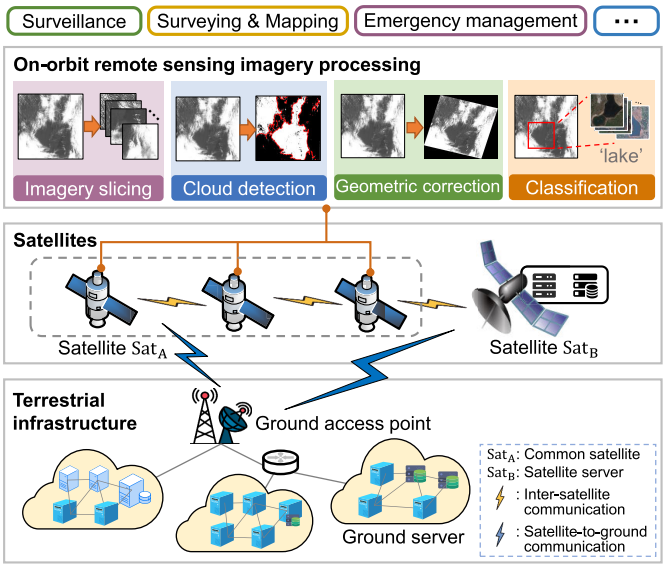



Fig. 1. SEC architecture for on-orbit remote sensing imagery processing.

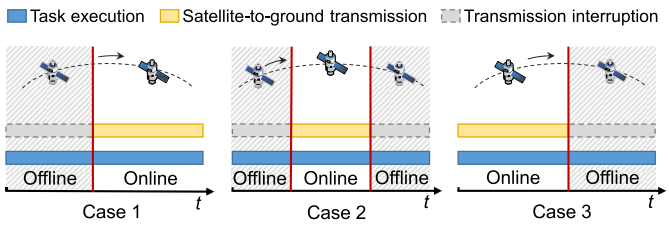



Fig. 2. Offline scenarios of the satellite with on-orbit task execution.

explore the related issues in SEC, especially facing on-orbit remote sensing imagery processing. The research problem proposed in this work requires to consider more factors, such as limited onboard resources, restricted power supply, and intermittent and bandwidth-constrained communication.

### C. Resource Scheduling Optimization Algorithm

At present, scheduling algorithms widely used by researchers can be classified into two distinct categories: metaheuristic approaches and knowledge-based learning techniques. For the former, genetic algorithm (GA), particle swarm optimization (PSO), ant colony optimization (ACO), and artificial bee colony algorithm (ABC) are classical and effective schemes used to solve scheduling problems [31]. Meanwhile, another alternative is to devise hybrid algorithms by fusing different types of metaheuristic strategies to sufficiently leverage their intrinsic advantages. For example, Attiya et al. [32] develop a hybrid swarm intelligence approach, incorporating the modified manta ray foraging optimization (MRFO) and salp swarm algorithm (SSA), to handle the problem of scheduling IoT tasks in cloud computing.

Deep reinforcement learning (DRL), as a novel end-to-end design paradigm for scheduling optimization, has triggered widespread interest in academia and industry. In [27], a parameterized deep Q-network (PDQN) is proposed to minimize the total latency through joint online service placement and computation resource allocation in 5G-enabled edge computing systems within stochastic environments. To reduce long-term operational expenditure, Gu et al. [24] present a novel DRL-based strategy that continuously explores states and adaptively makes decisions on service management and energy scheduling in edge computing. Another study [28] investigates the collaborative task computing and on-demand resource scheduling issue in VEC, proposing a multiresource orchestration framework that leverages asynchronous DRL to enhance system utility and ensure personalized service demand. In [33], a sophisticated multitask resource scheduling (MARS) framework is developed for dynamic MEC systems. The main ideology is designing a DRL-based approach to optimize resource allocation and energy cost. Ju et al. [25] use the DRL algorithm to design a VEC pioneering secure offloading and resource allocation (SORA) framework, which is validated to make the system more efficient and robust.

Despite that the existing work has designed metaheuristic algorithms to well solve scheduling problems, the inherent additional computational overhead rendered them inapplicable for real-time or near-real-time on-orbit services. Hence, some end-to-end algorithms based on DRL have been developed, but most single-agent DRL methods cannot applied to multiple satellite scenarios in SEC. Although a few multiagent DRL (MADRL) approaches have also been designed, they are not dedicated to addressing the SEC scheduling problem. Based on a review of the literature, few work investigate how to achieve efficient SEC based on-orbit imagery processing using end-to-end algorithm, which merits further investigation.

## III. SEC Resource Scheduling Model for On-Orbit Remote Sensing Imagery Processing

In this section, we first present the SEC architecture and its components, considering satellite-to-ground communication discontinuity and on-orbit remote sensing imagery processing missions. On this basis, the framework of the proposed SEC resource scheduling methodology is outlined. Next, an abstract task description graph is modeled from several typical on-orbit missions (i.e., imagery slicing, cloud detection, geometric correction, and classification). Finally, we formulate the SEC resource scheduling problem for on-orbit remote sensing imagery processing.

### A. SEC Architecture

*1) Computation:* The SEC computing devices consist of satellites and terrestrial infrastructure, as depicted in Fig. 1. Two types of satellites are included in SEC. One is the (1) common satellite (Sat$_A$), which generates on-orbit computation requirements and also provides some computing resources powered by onboard chips. The other is a (2) satellite server (Sat$_B$) dedicated to computing, namely, a special satellite with more hardware resources. As discussed above, due to the restriction of onboard resources, we use the (3) ground server to support on-orbit remote sensing imagery processing. Furthermore, we implement imagery slicing, cloud detection, geometric correction, and classification in this work to meet complicated on-orbit computing demands. Ultimately, this basic architecture can promote a variety of satellite services, such as ocean and farm surveillance, geographic surveying and mapping, and emergency management for disasters.

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*2) Communication:* Two transmission links including intersatellite and satellite-to-ground in SEC are primarily facilitated based on the laser technique. The intersatellite link supports communication among satellites, while the satellite-to-ground link transfers data from satellites to ground, and vice versa. SEC uses the ground access point (GAP) to collect and dispatch data. However, the connection between satellites and ground cannot be continuous since the mobility of satellites is in a wide range and the number of ground stations is limited. Consequently, the satellite loses the link with the ground when it roams outside the signal range of GAP.

*3) Online–Offline Mode:* The satellite faces a problem of limited connectable periods incurred by orbital characteristics. Therefore, an online–offline working mode is proposed to deal with the satellite-to-ground disconnection. Based on [34], [35], we define the online–offline in such a satellite constellation design scheme where satellites roam with the same orbital altitude and different orbital inclination. Moreover, a permanent intersatellite link without dynamic reconstruction is maintained based on the constellation configuration. In this scheme, all the satellites are in the offline state and unable to transmit data to ground stations when the constellation moves out of the GAP visible range. Next, we design three offline scenarios, as illustrated in Fig. 2. In case 1, satellite task execution starts during the offline period, which will switch to the online state. The mode in case 2 is triggered when the satellite connects with GAP during task execution and then disconnects from GAP before all the tasks are completed. Case 3 means that the satellite-to-ground transmission link interruption occurs during task execution, and the satellite remains offline until all the tasks are completed.

### B. Framework of the Proposed Methodology

This article proposes an SEC resource scheduling optimization scheme, or SECORS for short, to achieve efficient on-orbit remote sensing imagery processing. A comprehensive framework is illustrated in Fig. 3, where the resource scheduling model and multiagent reinforcement learning algorithm with the heuristic strategy are core components, denoted as SEC-RSM and SEC-MPH, respectively. The system commences with the generation of on-orbit remote sensing imagery processing tasks by satellites and uniformly abstracts them using a graph. Then, a mathematical formulation is defined for describing how to reasonably schedule SEC computational resources to facilitate collaborative task execution. Execution time and energy costs are calculated as the objectives. In the last step, we establish a training environment using SEC-RSM and devise an MADRL-based algorithm for the system, generating scheduling decisions to obtain faster on-orbit computational performance with lower energy cost.

### C. DAG for On-Orbit Remote Sensing Imagery Processing

We leverage the DAG to model on-orbit remote sensing imagery processing missions, as outlined in the upper portion of Fig. 3. In this work, four typical missions including imagery slicing, cloud detection, geometric correction, and classification are investigated as the case study. In particular, the first step is to construct the task chunks in DAG, which represent multiple tasks decomposed from remote sensing imagery processing programs [36]. Second, the directions of DAG are built to represent predecessors and successors among tasks, restricting the execution sequence in the program. On this basis, a dedicated DAG for satellite applications can be defined. Below is the detailed DAG construction process for each mission.

*1) Imagery Slicing:* Imagery slicing provides the function of dividing a large-scale remote sensing image into multiple smaller slices, which is developed for subsequent parallel processing. As shown in Fig. 4, the program initially loads an input remote sensing image through the geospatial data abstraction library (GDAL) [37]. With the input parameter of slice granularity, the number of pixels and offset of each slice are calculated and thereafter the objective image is segmented into slices with original values, data types, and band information. Having dissected the imagery slicing implementation, we design the related task graph, which is illustrated using purple nodes in DAG in Fig. 3. The whole process is denoted using two tasks, which are data loading, and parameter calculation and image segmentation, respectively.

*2) Cloud Detection:* Cloud detection is implemented using Fmask algorithm [38], [39], whose program flow is demonstrated in Fig. 4. Several key information are obtained through GDAL tool, spanning the top of atmosphere reflectances for bands 1–5 and 7, and band 6 brightness temperature. Then, the Fmask method uses several physical properties of cloud and cloud shadow to recognize the potential cloud layer and cloud shadow layer. During this process, cloud probability results for two passes of the potential cloud layer are estimated through spectral tests. Besides, the morphological transformation with band 4 reflectance is performed for the potential cloud shadow layer. The last step is to match the segmented potential cloud layer and geometric relationships with the potential cloud shadow layer, therefore creating the cloud mask. Based on the algorithmic analysis, we establish DAG for cloud detection, which is depicted as the blue marked nodes of the task model in Fig. 3. Fmask-based cloud probability estimation manifests an intrinsic parallel property of single instruction, multiple data (SIMD). This operation is equally divided into many task nodes for simultaneous execution, where the number of task nodes is determined by the image scale. The last node means generating the cloud layer outcome.

*3) Geometric Correction:* As elaborated in Fig. 4, the program of geometric correction starts with the input data from GDAL, and then performs a resampling operation based on rational function model (RFM) [40]. This involves rational polynomial coefficients (RPCs) to compute the geodetic coordinates for every pixel [41]. Next, the gray value is calculated by means of nearest neighbor interpolation. After that, merging all the imagery slices is the last step to create a new image with corrected pixels. With regard to DAG in Fig. 3, we leverage green marked task nodes to describe geometric correction. It can be found that RFM-based resampling operation has explicit SIMD parallelism, and thus is modeled with multiple concurrent tasks. Another task is established to indicate that

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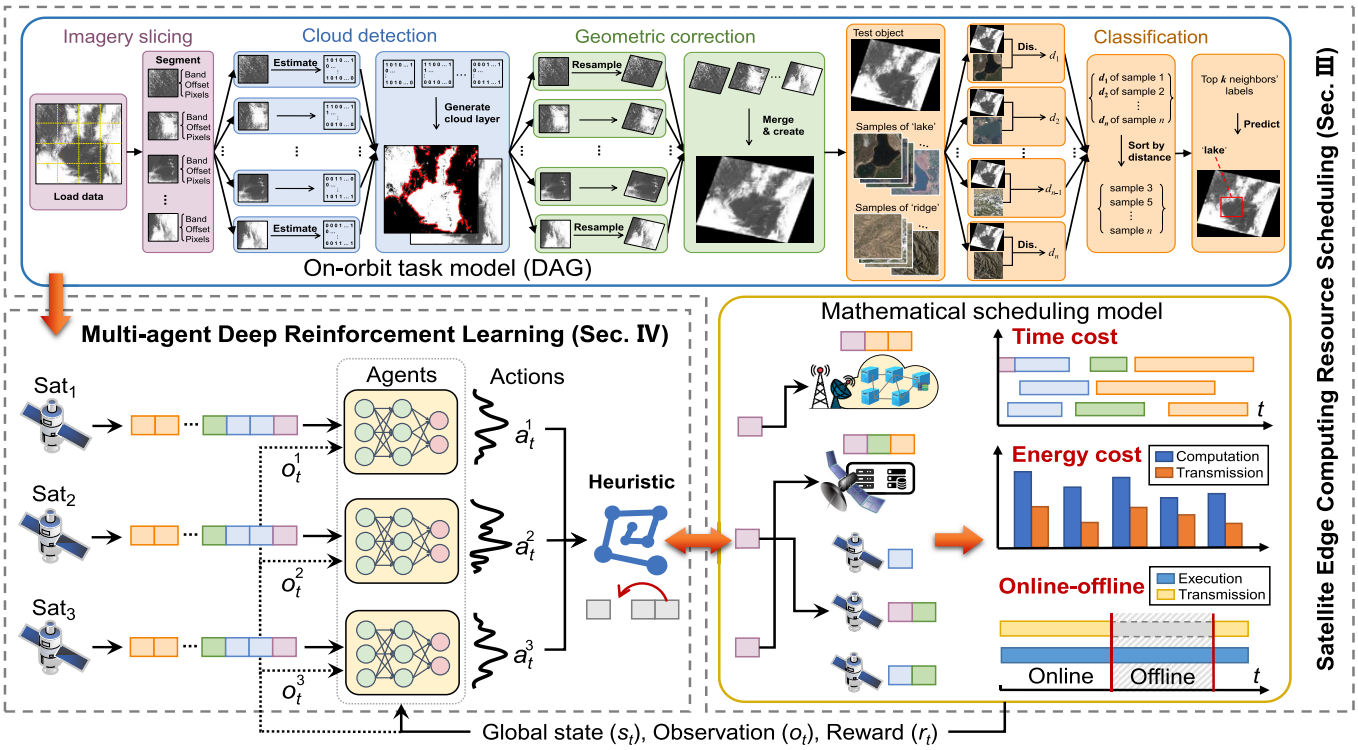



Fig. 3. Framework of SEC resource scheduling optimization for on-orbit remote sensing imagery processing.

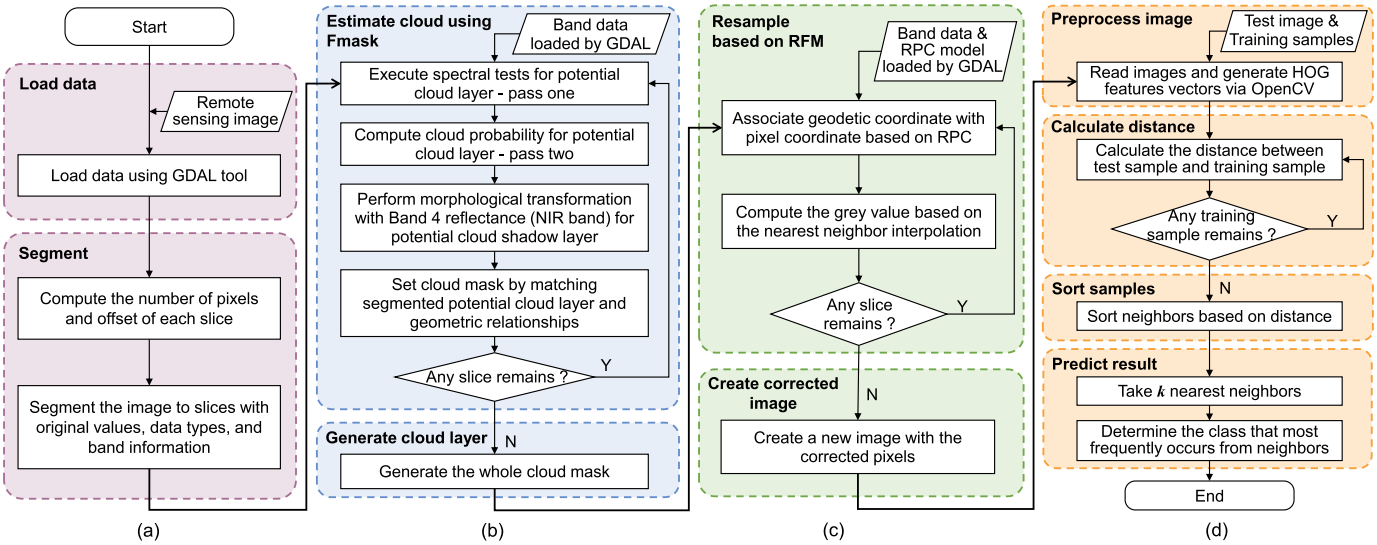



Fig. 4. Algorithmic flowchart of on-orbit missions, i.e., imagery slicing, cloud detection, geometric correction, and classification. (a) Imagery slicing. (b) Cloud detection. (c) Geometric correction. (d) Classification.

the following operation is to merge and generate the geometrically corrected image.

*4) Classification:* We leverage the $k$-nearest neighbor (KNN) algorithm [42] to classify remote sensing images, the program flow of which is demonstrated in Fig. 4. During the procedure, an open-source computer vision (OpenCV) library is imported to load image samples and then convert them into feature descriptors using histogram of oriented gradient (HOG) [43]. After obtaining the samples, the following steps are executed, which are calculating distances between the test object and training samples, sorting distances with related classes, and determining classification results from KNNs. Meanwhile, we establish a corresponding task model (depicted as the orange area of DAG in Fig. 3). The operation of data loading and feature description is designed as a task. Subsequently, multiple parallel tasks are constructed for Euclidean distance calculation between the test object and all train samples (or neighbors). Each task processes one neighbor or several neighbors, which depends on how many parallel tasks are set up. The next two task nodes indicate that all neighbors are sorted according to their distances with the test object, and finally, the most frequently occurring class from $k$ closest neighbors serves as the inference result.

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### D. Mathematical Resource Scheduling Problem Formulation

In this section, we describe the SEC resource scheduling problem, establishing a mathematical model (SEC-RSM). Generally, it serves to calculate time and energy costs for a scheduling solution required to adhere to sequence constraints from DAG. On one hand, time cost consists of satellite task execution and data transmission after resource scheduling. If there exists a sequence relationship (or the directed edge in DAG) between two tasks, the successor one can be executed only after the predecessor one has been accomplished. Data transmission is incurred if two tasks with sequence constraints are allocated with different devices, which is supported by intersatellite and satellite-to-ground links. On the other hand, considering the extremely constrained power supply on satellite equipment, we define a fine-grained energy calculation formulation for SEC-RSM. The energy is mainly consumed by each device and laser communication facility in SEC during task execution and data transmission. All the notations commonly used in this section are listed in the Nomenclature.

On the basis of the above-mentioned description and analysis, the research problem is formulated as below

$$\min C^{\mathrm{T}}, C^{\mathrm{T}}. \tag{1}$$

Subject to

$$\sum_{i\in\mathcal{T}\cup 0'} x_{0i}^{k} = 1 \quad \forall k \in \mathcal{K} \tag{2}$$

$$\sum_{i\in\mathcal{T}\cup 0} x_{i0'}^{k} = 1 \quad \forall k \in \mathcal{K} \tag{3}$$

$$\sum_{j\in\mathcal{T}^{+},j\neq i} x_{ij}^{k} = \sum_{j\in\mathcal{T}^{+},j\neq i} x_{ji}^{k} \quad \forall i \in \mathcal{T},\ k \in \mathcal{K} \tag{4}$$

$$\sum_{k\in\mathcal{K},j\in\mathcal{T}^{+}} x_{ij}^{k} = 1 \quad \forall i \in \mathcal{T} \tag{5}$$

$$t_i^{\mathrm{end}} \geq t_i^{\mathrm{start}} + \sum_{j\in\mathcal{T}^{+},k\in\mathcal{K}} w_i^{k} x_{ij}^{k} \quad \forall i \in \mathcal{T} \tag{6}$$

$$t_j^{\mathrm{start}} \geq t_i^{\mathrm{end}} - M\left(1 - \sum_{k\in\mathcal{K}} x_{ij}^{k}\right) \quad \forall i, j \in \mathcal{T} \tag{7}$$

$$t_i^{\mathrm{tran_start}} \geq t_i^{\mathrm{end}} \quad \forall i \in \mathcal{T} \tag{8}$$

$$t_i^{\mathrm{tran_end}} \geq t_i^{\mathrm{tran_start}} + d_{ij}^{kl} - M\left(1 - y_{ij}^{kl}\right) \quad \forall (i, j) \in \mathcal{P},\ k, l \in \mathcal{K} \tag{9}$$

$$t_i^{\mathrm{tran_end}} \leq t^{\mathrm{off_start}} + M\left(1 - y_{ij}^{kl}\right) + M z_i \quad \forall (i, j) \in \mathcal{P}, (k, l) \in \mathcal{Q} \tag{10}$$

$$t_i^{\mathrm{tran_start}} \geq t^{\mathrm{off_end}} - M\left(1 - y_{ij}^{kl}\right) - M(1 - z_i) \quad \forall (i, j) \in \mathcal{P},\ (k, l) \in \mathcal{Q} \tag{11}$$

$$y_{ij}^{kl} \leq \sum_{i'\in\mathcal{T}\cup 0'} x_{ii'}^{k} \quad \forall i, j \in \mathcal{T}^{+},\ k, l \in \mathcal{K} \tag{12}$$

$$y_{ij}^{kl} \leq \sum_{j'\in\mathcal{T}\cup 0'} x_{jj'}^{k} \quad \forall i, j \in \mathcal{T}^{+}, k, l \in \mathcal{K} \tag{13}$$

$$y_{ij}^{kl} \geq \sum_{i'\in\mathcal{T}\cup\{0'\}} x_{ii'}^{k} + \sum_{j'\in\mathcal{T}\cup\{0'\}} x_{jj'}^{k} - 1, \quad \forall i, j \in \mathcal{T}^{+},\ k, l \in \mathcal{K} \tag{14}$$

$$t_i^{\mathrm{start}}, t_i^{\mathrm{end}}, t_i^{\mathrm{tran_start}}, t_i^{\mathrm{tran_end}} \geq 0 \quad \forall i \in \mathcal{T} \tag{15}$$

$$C^{\mathrm{T}} \geq t_i^{\mathrm{end}} \quad \forall i \in \mathcal{T} \tag{16}$$

$$C^{\mathrm{E}} = \sum_{i,j\in\mathcal{T}^{+},k\in\mathcal{S}} x_{ij}^{k} w_i^{k} p_k + \sum_{(i,j)\in\mathcal{P},k,l\in\mathcal{K}} y_{ij}^{kl} d_{ij}^{kl} p^{\mathrm{tran}} \tag{17}$$

$$x_{ij}^{k} \in \{0,\ 1\} \quad \forall i, j \in \mathcal{T}^{+},\ k \in \mathcal{K} \tag{18}$$

$$y_{ij}^{kl}, z_i \in \{0, 1\} \quad \forall i, j \in \mathcal{T},\ k, l \in \mathcal{K}. \tag{19}$$

For the model, (1) presents the optimization objectives concerning time and energy overhead. Constraints (2) and (3) identify the requisite entry and exit tasks on each computing device. Constraint (4) denotes two consecutively executed tasks, that is, no other tasks are carried out between these two tasks, while (5) restricts one task can just be handled by one device. Then, we define (6) and (7) to describe the relationship between the start and end time for the task, as well as the execution sequence limitation of the next task, respectively. Equations (8) and (9) are formulated to calculate the data transmission time in the scenario without being influenced by offline. An intrinsic condition is that the transmission only occurs among the predecessor and successor tasks. Similarly, if two dependent tasks are scheduled to the offline satellite device and terrestrial device, respectively, constraints (10) and (11) are used to compute the transmission time affected by the offline state. The interrelationship of $x_{ij}^{k}$ and $y_{ij}^{kl}$ is defined through (12)–(14). Equation (15) serves to restrict the values of intermediate variables, thereby ensuring the solvability of the model. Equations (16) and (17) then provide computation methods for two objectives. The final two constraints define the range of values for decision variables.

## IV. MAPPO and Heuristic-Based SEC Resource Scheduling Algorithm

We develop an MADRL-based scheduling approach called SEC-MPH with SEC-RSM as the environment. The design philosophy is based on the MDP analysis of the proposed SEC resource scheduling problem. To be specific, this work uses SEC-RSM as the training environment and redesigns the action, observation, and reward function for the multiagent system. Then, we leverage MAPPO as the updating method to train agents in a cooperative way. Besides, MAPPO only makes the decision on how to schedule resources to tasks, and it could not determine the execution sequence for several tasks assigned to one device. Thus, a scheduling heuristic strategy called EFT is used to determine the task sequence.

### A. MDP Formulation

In our defined scenario, the constellation in SEC compromises common satellites and a satellite server dedicated to computing. Common satellites generate on-orbit computation tasks modeled as DAG. In light of task sequence constraints in DAG, we first leverage the topological sorting algorithm to construct an initial task queue for each satellite. All the devices in SEC are denoted by a set $Q = \{q_1, q_2, \ldots, q_n, q_{n+1}, q_{n+2}\}$, where $e = 1, 2, \ldots, n$ and $e' = 1, 2, \ldots, n+2$. Here, $q_e$ represents the common satellites (generating on-orbit computation). $q_{e'}$ represents the device that provides computational resources, including $q_e$, the ground server $q_{n+1}$, and the satellite server $q_{n+2}$. Considering multiple

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satellites, the research problem requires a multiagent decision system. Thus, we use a partially observable MDP (POMDP) to analyze the research problem. POMDP can be defined as a seven-tuple $\langle S, A, O, R, P, n, \gamma \rangle$. $S$ represents the set of global environmental states, where $s_t$ denotes the state at time $t$. $A$ represents the set of all actions that agents can take, where $a_t$ denotes the joint actions of all agents at time $t$ and $a_e^t$ denotes the action of agent $e$. It can be noted that each agent $e$ is associated with a satellite $q_e$. $O$ represents the set of local observations of all agents, where $o_e^t$ denotes the local observations of agent $e$. The state transfer function $P_{s_t,s_{t+1}}^{a_t} = \mathbb{P}[s_{t+1} \mid s_t, a_t]$ represents the probability of the system transitioning from the state $s_t$ to the next state $s_{t+1}$, after executing action $a_t$. $R$ represents the set of rewards obtained after all agents execute actions. $n$ represents that there are $n$ agents, and $e = 1, 2, \ldots, n$. $\gamma$ denotes the reward discount factor.

*1) State and Observation Spaces:* To better describe the state of each SEC device, we define the state space as

$$s_t = \{NQ, CQ, CS, F\} \in S \tag{20}$$

where $NQ$ represents the set of the number of tasks handled by each device, $CQ$ represents the set of the completion time of each device, $CS$ represents the set of the completion time of the current task generated by each satellite, and $F$ represents the set used to describe whether the satellite is in offline status. We use $cs_e \in CS$ and $f_e \in F$ to denote the task completion time and offline status of satellite $e$, respectively. According to the global state, the observation space of each agent can also be defined as the following equation:

$$o_t^e = \{ND, CQ, \{cs_e\}, \{f_e\}\}. \tag{21}$$

*2) Action Space:* At a time, each agent executes an action on the related satellite, deciding which computing device is scheduled to process the first task in the queue established from DAG. The action space is defined below

$$a_t^e = (\varphi_1, \varphi_2, \ldots, \varphi_{e'}, \ldots, \varphi_{n+2}) \tag{22}$$

$$a_t = \left\{a_t^1, a_t^2, \ldots, a_t^e, \ldots, a_t^n\right\} \in A \tag{23}$$

where $a_t^e$ is the action of agent $e$, and $\varphi_1, \varphi_2, \ldots, \varphi_{n+2}$ are valued in one-hot encoding format. For example, $\varphi_{e'} = 1$ with other values in $a_t^e$ being 0 indicates that device $q_{e'}$ is allocated to deal with the current task in satellite $e$. Equation (23) represents the joint action for all the satellites.

*3) Reward:* This study designs a hybrid reward function, incorporating a global reward and private rewards for each agent. The objective of our SEC-RSM is to minimize the total time ($C^T$) and energy ($C^T$) cost, as presented in (16) and (17). Thus, we define the reward function in each time step based on these two indicators

$$r_t^e = -cs_e \tag{24}$$

$$r_t^{\mathrm{G}} = -\left[\lambda_1 \beta C^T + \lambda_2 (1-\beta) C^T\right] \tag{25}$$

$$r_t = \frac{1}{n} \sum_{e=1}^{n} \omega r_t^e + (1-\omega) r_t^{\mathrm{G}} \tag{26}$$

where $r_t^e$ means the private reward of agent $e$, and the calculation is based on its observation value $cs_e$. $r_t^{\mathrm{G}}$ refers to the global reward, which can be computed using the results of SEC-RSM. In this equation, $\lambda_1$ and $\lambda_2$ are normalization coefficients used to combine time and energy cost, while $\beta$ is the time–energy weight factor. $r_t$ indicates the average reward after taking action $a_t$ in state $s_t$, and $\omega$ is the global–private weight coefficient.

### *B. Proposed Algorithm Based on MAPPO and Heuristic (*SEC-MPH*)*

In MAPPO [44], [45], each agent consists of two independent networks: policy network $\pi_\theta$, parameterized by $\theta$, and value network $V_\phi$, parameterized by $\phi$. The policy and value networks are also called the actor and critic. All the agents are constructed using the same actor–critic structure. The policy $\pi$ of the agent is the mapping probability from local observation to action. At any time $t$, the state of the system is $s_t$, and agent $e$ obtains its observation $o_t^e$ according to the observation $O$ and then selects the action $a_t^e$ through $\pi_\theta(a \mid s)$. The goal of the reinforcement learning algorithm is to optimize $\pi_\theta(a \mid s)$ by maximizing the cumulative discounted reward $R_t$ with discount factor $\gamma \in (0, 1)$, calculated by (27). In addition, the critic network $V_\phi$ is used to assess the action $a_e^t$ taken by the agent $e$ in the environmental state $s_t$

$$R_t = r_t + \gamma^1 r_{t+1} + \gamma^2 r_{t+2} + \cdots = \sum_{t'=0}^{\infty} \gamma^{t'} r_{t+t'}. \tag{27}$$

MAPPO uses an advantage function to evaluate the advantage of taking a particular action in its current state, indicating that an action may gain higher or lower rewards compared with other actions. To overcome the nonconvergence during multiagent training process, the generalized advantage estimation (GAE) is adopted. Its calculation is given by

$$\hat{A}_{t,e}^{\mathrm{GAE}} = \sum_{t'=0}^{\infty} (\gamma \varrho)^{t'} \left[r_{t+t'} + \gamma V_\phi^e(s_{t+t'+1}) - V_\phi^e(s_{t+t'})\right] \tag{28}$$

where $\varrho \in (0, 1)$ is the GAE discount factor. Besides, MAPPO uses the clipping function with a ratio $\epsilon$ to limit the magnitude of each $\theta$ update. On this basis, an updating strategy for the actor network is obtained by the following equation:

$$L_\theta^{\mathrm{CLIP}} = \frac{1}{Bn} \sum_{b=1}^{B} \sum_{e=1}^{n} \min \times \left[\frac{\pi_\theta(a_b^e \mid o_b^e)}{\pi_{\theta_{\mathrm{old}}}(a_b^e \mid o_b^e)} \hat{A}_{b,e}^{\mathrm{GAE}}, \mathrm{clip} \times \left(\frac{\pi_\theta(a_b^e \mid o_b^e)}{\pi_{\theta_{\mathrm{old}}}(a_b^e \mid o_b^e)}, 1-\epsilon, 1+\epsilon\right) \hat{A}_{b,e}^{\mathrm{GAE}}\right] \tag{29}$$

where $B$ is the number of batch processes, and $n$ is the number of satellites (or corresponding agents). Accordingly, the loss function $L^{\mathrm{CLIP}}(\phi)$ of the value network is given as

$$L_\phi^{\mathrm{CLIP}} = \frac{1}{Bn} \sum_{b=1}^{B} \sum_{e=1}^{n} \max \times \left[\left(V_\phi^e(s_b) - R_b\right)^2, \left(\mathrm{clip}\left(V_\phi^e(s_b), V_{\phi_{\mathrm{old}}}^e(s_b) - \epsilon, V_{\phi_{\mathrm{old}}}^e(s_b) + \epsilon\right) - R_b\right)^2\right]. \tag{30}$$

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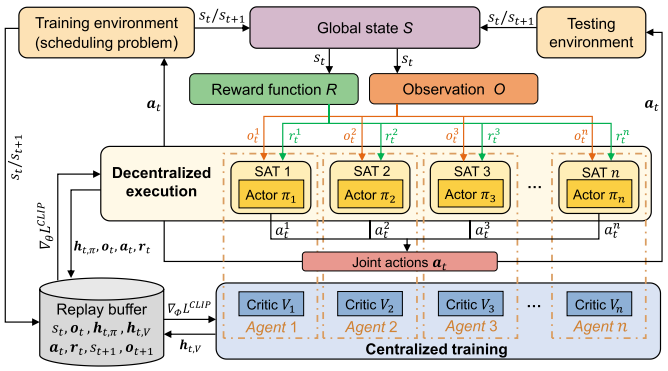



Fig. 5. Training and execution process of SEC-MPH.

EFT [46], a classical heuristic scheduling strategy, is another key component in our proposed algorithm. In such a scenario when multiple tasks are assigned to be executed on the same device, an inappropriate execution sequence will lead to a proliferation of time overhead. To cope with this issue, we design a method based on the EFT mechanism, which is triggered after MAPPO agents execute actions. Specifically, EFT chooses these devices scheduled with more than one task and computes the completion times of all tasks for each device. Then, it sorts tasks in ascending order based on the completion time, outputting a sequence. Correspondingly, locally optimal task execution solutions are obtained by the above operations.

In this article, we use a centralized training and decentralized execution (CTDE) framework to implement SEC-MPH. As illustrated in Fig. 5, initially, each agent observes the current state from the global state generated by the environment. When receiving observation results, the actor network makes a decision and outputs a scheduling action for the current training step. The joint actions of all agents are input into the environment constructed by SEC-RSM, and then a new state information will be generated. Meanwhile, the EFT heuristic strategy is triggered to sequence the tasks scheduled with the same computing device. The reward function computes an immediate reward and the critic network outputs a related evaluation for this action. Subsequently, an experience replay buffer is set to store several crucial information such as state, observation, action, reward, and hidden parameters of recurrent neural network (RNN) for actor and critic components, denoted as $[s_t, \boldsymbol{o}_t, \boldsymbol{h}_{t,\pi}, \boldsymbol{h}_{t,V}, \boldsymbol{a}_t, \boldsymbol{r}_t, s_{t+1}, \boldsymbol{o}_{t+1}]$. Finally, a certain number of experiences are sampled from this buffer to calculate the gradient and then the parameters of the actor and critic networks are updated through the Adam optimizer. A detailed algorithm flow is provided in Algorithm IV-B.

### C. Computational Complexity Analysis

In this study, we analyze the computational complexity of the proposed algorithm SEC-MPH on its execution. First, an RNN is leveraged as the actor network responsible for actions in a single agent. The RNN's running complexity can be identified with $O(nd^2)$ [47], where $n$ is the length of the input sequence and $d$ is the dimension of each input element. Moreover, the algorithm calculates the relevant reward by calling the proposed problem model SEC-RSM to obtain the time and energy costs. This involves scheduling $(N+2)$ SEC devices to execute current tasks on $N$ satellites, resulting in a computational complexity of $O[N(N+2)]$, which can be further simplified to $O(N^2)$. Next, for multiple agents, the aforementioned process can be executed simultaneously using parallel programming methods. Therefore, the complexity of

**Algorithm 1** SEC-MPH

**Input:** Learning rates of actor and critic networks $lr_1, lr_2$, discount factors $\gamma, \varrho$, coefficients $\lambda_1, \lambda_2, \beta, \omega$, clip ratio $\epsilon$, sampled length $\tau$, maximum episode $EP$ batch size $B$, chunk length $L$, and gradient update times $K$.
**Output:** Cumulative reward, time, and energy cost.

Load model SEC-RSM and obtain $n$ (the number of satellites) and $T$ (the maximum steps in an episode)
Initialize $\theta$ of actor $\pi$, and $\phi$ of critic $V$
Transfer the DAG in each satellite to the task queue, using the topological sorting approach
**while** the maximum episode $EP$ is not required **do**
  Set a replay buffer $D = \{\}$
  **for** $b = 1$ to $B$ **do**
    $\tau =$ [ ] empty list
    Initialize environment (reset model SEC-RSM)
    Initialize $h^1_{0,\pi}, \ldots, h^n_{0,\pi}$ for actor RNN
    Initialize $h^1_{0,V}, \ldots, h^n_{0,V}$ for critic RNN
    **for** $t = 1$ to $T$ **do**
      **for** all satellite agents $e$ **do**
        $p^e_t, h^e_{t,\pi} = \pi(a^e_t | o^e_t, h^e_{t-1,\pi}; \theta)$
        $a^e_t \sim p^e_t$
        $v^e_t, h^e_{t,V} = V(s_t, h^e_{t-1,V}; \phi)$
      **end for**
      Execute joint action $a^1_t, a^2_t, \ldots, a^n_t$
      Sequence tasks using the heuristic of EFT
      Observe $\boldsymbol{r}_t, s_{t+1}, \boldsymbol{o}_{t+1}$
      $\tau$ += $[s_t, \boldsymbol{o}_t, \boldsymbol{h}_{t,\pi}, \boldsymbol{h}_{t,V}, \boldsymbol{a}_t, \boldsymbol{r}_t, s_{t+1}, \boldsymbol{o}_{t+1}]$
    **end for**
  **end for**
  Compute advantage estimate $\hat{A}^{\mathrm{GAE}}$ and cumulative discounted reward $R$ on $\tau$
  Split trajectory $\tau$ into chunks of length $L$
  **for** $\mathscr{L} = 0, \ldots, T//L$ **do**
    $D = D \cup \{\tau[\mathscr{L} : \mathscr{L} + L], \hat{A}^{\mathrm{GAE}}[\mathscr{L} : \mathscr{L} + L], R[\mathscr{L} : \mathscr{L} + L]\}$
  **end for**
  **for** mini-batch $\mathscr{K} = 1$ to $K$ **do**
    $mb \leftarrow$ random mini-batch from $D$ with all agent data
    **for** each data chunk $c$ in the mini-batch $mb$ **do**
      update RNN hidden states for $\pi$ and $V$ from first hidden state in data chunk
    **end for**
  **end for**
  Calculate the gradient $\nabla_\theta L^{\mathrm{CLIP}}$ and update $\theta$ on $L^{\mathrm{CLIP}}_\theta$ with data $mb$ and learning rate $lr_1$
  Calculate the gradient $\nabla_\phi L^{\mathrm{CLIP}}$ and update $\phi$ on $L^{\mathrm{CLIP}}_\phi$ with data $mb$ and learning rate $lr_2$
**end while**

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TABLE I
PARAMETER SETTING OF SEC-MPH

| Parameter | Value |
|---|---|
| Learning rates $lr_1, lr_2$ | 0.0001, 0.0001 |
| Discount factors $\gamma, \varrho$ | 0.99, 0.95 |
| Coefficients $\lambda_1, \lambda_2, \beta, \omega$ | 1, 0.01, 0.5, 0.01 |
| Clip ratio $\epsilon$ | 0.2 |
| Sampled length $\tau$ | 10 |
| Maximum episode $EP$ | 10000 |
| Batch size $B$ | 8 |
| Chunk length $L$ | 10 |
| Gadient update times $K$ | 10 |

a scheduling decision containing the action execution and reward calculation for multiagent is $O(nd^2 + N^2)$. Finally, since all the tasks on the satellites need to be allocated, incurring $T$ times of scheduling decision, the overall computational complexity of the algorithm is $O[T(nd^2 + N^2)]$.

## V. EXPERIMENTAL RESULTS AND ANALYSIS

In this section, we first build a simulation SEC system as the experimental platform, which involves the installation of hardware devices and configuration of the software running environment. Next, a dedicated dataset is designed by selecting remote sensing imagery, constructing DAG test instances, and recording statics on imagery processing execution time and transmission cost. Finally, a range of numerical experiments are carried out to validate the performance of our proposed SECORS. Specifically, the statistical data and DAGs are used as input parameters to be fed into SEC-RSM and then SEC-RSM is solved by SEC-MPH. Table I lists the detailed setting of parameters used in SEC-MPH. For comparative analysis, we use several satellite remote imagery processing schemes, including nonoptimization, SEC based on MADRL, and SEC based on evolutionary multiobjective optimization (EMO). An elaborate description is outlined as follows.

1) *Baseline:* It is a traditional satellite data processing mode, that is, each satellite is responsible for its own task and there is no SEC framework with resource scheduling optimization for the constellation.
2) *MADRL:* We implement a variety of MADRL algorithms to solve SEC-RSM, i.e., counterfactual multiagent policy gradients (COMAs) [48], value-decomposition actor–critic (VDAC) [49], and lazy agents avoidance through influencing external states (LAIESs) [50].
3) *EMO:* To compare various types of solution methods addressing SEC-RSM, we also develop two typical EMO algorithms with reference to [51] and [52], including nondominated sorting GA (NSGA-II) and multiobjective evolutionary algorithm based on decomposition (MOEA/D). Here, two key parameters, i.e., the population size and maximum iteration step, are set to 20 and 200, respectively.

### A. Experimental Setup

*1) SEC Simulation Platform:* The established experimental platform is equipped with various devices including servers and edge computers to simulate SEC on-orbit computation, as shown in Fig. 6. We deploy the ground server with a CPU + GPU setup; another CPU + FPGA setup is used to construct low-power satellite edge servers; and for common on-orbit satellites, the embedded CPU is used as the onboard computing unit. In addition, this work uses a local area network (LAN) to interconnect various devices, facilitating data transmission. As for software configuration, the operating system (OS) of SEC is Ubuntu, upon which a framework called open computing language (OpenCL) is installed to support universal programming for heterogeneous processors. Meanwhile, the running environment supporting C++ and Python is also assembled in this system to implement the proposed SEC-RSM and SEC-MPH, respectively. According to the core configurations released on official vendor sites, all the setting parameters concerning hardware and software used in this article are listed in Tables II and III. At this point, we have completely established a simulated SEC platform incorporating our SECORS technique.

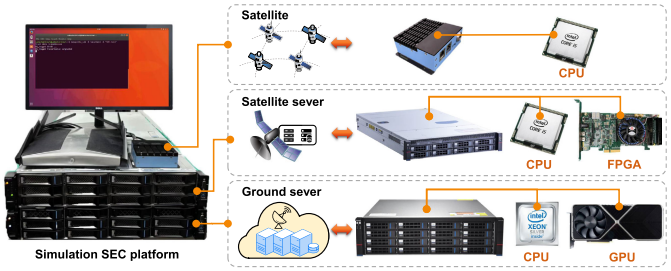



Fig. 6. Simulation SEC platform.

TABLE II
HARDWARE CONFIGURATION OF THE SIMULATION PLATFORM

| SEC hardware | Quantity | Computing resource | Trans. rate | Power |
|---|---|---|---|---|
| Satellite | {3,5,10,15} | CPU: Intel i5-1035G4 | / | 15 W |
| Satellite server | 1 | CPU: Intel i5-1035G4<br>FPGA: Intel Cyclone V GT | / | 25 W |
| Ground server | 1 | CPU: Intel Xeon Silver 4201<br>GPU: NVIDIA RTX 3090 | / | 220 W |
| Inter-satellite link | / | / | 10 Gbps | 5 W |
| Satellite-to-ground link | / | / | 1 Gbps | 5 W |

TABLE III
SOFTWARE CONFIGURATION OF THE SIMULATION PLATFORM

| SEC software | Version |
|---|---|
| Operating system | Ubuntu 14.04 |
| Running env. for SEC-RSM | C/C++ 8.1 & OpenCL 1.2 |
| Running env. for SEC-MPH | Python 3.8 |

*2) Dataset:* In terms of remote sensing images used in on-orbit computation, including imagery slicing, cloud detection, geometric correction, and classification, we download

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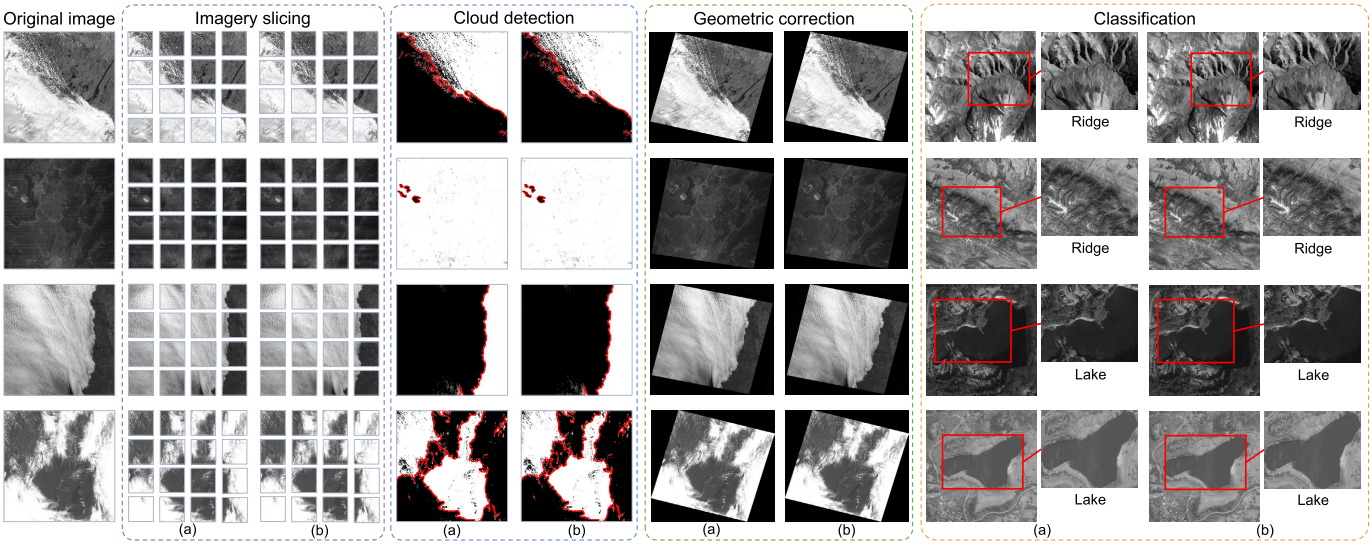



Fig. 7. Results of remote sensing imagery processing through different programs. (a) Baseline program using C++. (b) SEC program using OpenCL.

TABLE IV
NUMBER OF TASKS IN DAGS WITH DIFFERENT IMAGE SCALES

| On-orbit mission | Task | 256 | 512 | 1024 | 2048 | 4096 | 8192 |
|---|---|---|---|---|---|---|---|
| Imagery slicing | IS-LAD | 1 | 1 | 1 | 1 | 1 | 1 |
| | IS-SEG | 1 | 1 | 1 | 1 | 1 | 1 |
| Cloud detection | CD-FMK | 4 | 8 | 16 | 32 | 64 | 128 |
| | CD-GEN | 1 | 1 | 1 | 1 | 1 | 1 |
| Geometric correction | GC-RFM | 4 | 8 | 16 | 32 | 64 | 128 |
| | GC-MEG | 1 | 1 | 1 | 1 | 1 | 1 |
| Classification | CF-PRE | 1 | 1 | 1 | 1 | 1 | 1 |
| | CF-CAL | 8 | 8 | 16 | 16 | 32 | 64 |
| | CF-SRT | 1 | 1 | 1 | 1 | 1 | 1 |
| | CF-RES | 1 | 1 | 1 | 1 | 1 | 1 |
| **Total tasks in DAG** | | **23** | **31** | **55** | **87** | **167** | **327** |

Note: 256 means 256×256, and other values follow this pattern. IS-LAD and -SEG indicate *imagery slicing*; CD-FMK and -GEN indicate *cloud detection*; GC-RFM and -MEG indicate *geometric correction*; CF-PRE, -CAL, -SRT, and -RES indicate *classification*.

them from https://www.usgs.gov/landsat-missions and https://github.com/zhangbin0917/Awesome-Remote-Sensing-Dataset, respectively. For the first link, some generic Landsat remote sensing pictures with multiple bands can be used for verifying imagery slicing, cloud detection, and geometric correction. The next website provides samples from Google Earth imagery, which is used here for executing classification functions. We resize images into six scales, namely, 256 × 256, 512 × 512, 1024 × 1024, 2048 × 2048, 4096 × 4096, and 8192 × 8192. On top of that, we construct various DAGs as on-orbit missions according to different image scales, which is based on the SIMD parallel feature analyzed in Section III-C. Here, different task nodes in DAG are denoted with abbreviations. IS-LAD and -SEG indicate loading data and segmenting imagery in *imagery slicing*; CD-FMK and -GEN indicate Fmask cloud estimation and generating cloud layer in *cloud detection*; GC-RFM and -MEG indicate resampling based on RFM and creating the corrected image in *geometric correction*; and CF-PRE, -CAL, -SRT, and -RES indicate preprocessing image, calculating Euclidean distance, sorting samples, and predicting the final result in *classification*. As listed in Table IV, DAGs with different total tasks are set according to image scales.

### B. Correctness Validation of Imagery Processing

Prior to commencing performance validation, this article identifies whether the proposed method can work correctly by running the programs of baseline (implemented by C++) and SECORS (implemented by OpenCL). Then the results of remote sensing imagery processing, i.e., imagery slicing, cloud detection, geometric correction, and classification, are compared, which are presented in Fig. 7. Columns Fig. 7(a) and (b) demonstrate the outcomes generated by the baseline program and SEC program, respectively. It can be noted that two programs create the same slices after executing the imagery slicing function. Regarding cloud detection, the Fmask method implemented in two schemes demonstrates a promising capability to generate clear cloud layers, the main areas of which are delineated by red lines. Similarly, the programs designed for geometric correction present consistent corrected images. For classification, this work establishes two categories (i.e., lake and ridge), each comprising 32 training samples. A total of 100 images were selected for testing, and several results are presented herein. Overall, the average accuracy of image classification using the KNN + HOG method is 73.00%.

Besides, by verifying the correctness, we also record the execution time (milliseconds) of each task in DAG and the data transmission cost (milliseconds) between two tasks assigned to different SEC devices. All the experimental results are listed in Tables V and VI, respectively. In the forthcoming context, we conduct evaluation experiments of the proposed SECORS based on the statistical execution cost and transmission cost, as well as DAGs modeled with varying image scales defined in Section V-A2.

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TABLE V

EXECUTION TIME COST (MS) OF IMAGERY PROCESSING TASKS RUNNING ON SEC PLATFORM

| On-orbit task | 256×256 | | | 512×512 | | | 1024×1024 | | | 2048×2048 | | | 4096×4096 | | | 8192×8192 | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | SAT | SAS | GND | SAT | SAS | GND | SAT | SAS | GND | SAT | SAS | GND | SAT | SAS | GND | SAT | SAS | GND |
| IS-LAD | 0.2947 | 0.2947 | 0.1122 | 0.3022 | 0.3022 | 0.1148 | 0.3561 | 0.3561 | 0.1354 | 0.3774 | 0.3774 | 0.1435 | 0.4863 | 0.4863 | 0.1849 | 0.5491 | 0.5491 | 0.2088 |
| IS-SEG | 0.6247 | 0.2865 | 0.1011 | 1.0398 | 0.4228 | 0.2804 | 1.8028 | 0.8915 | 0.5048 | 4.2396 | 2.0986 | 1.1102 | 6.8529 | 3.7902 | 2.1552 | 10.0348 | 6.2753 | 3.7204 |
| CD-FMK | 0.9724 | 0.4166 | 0.1881 | 1.6486 | 0.7742 | 0.3974 | 3.3308 | 1.4986 | 0.8094 | 6.5196 | 2.9370 | 1.6112 | 13.0392 | 5.8740 | 3.2225 | 26.4696 | 11.7563 | 6.3483 |
| CD-GEN | 0.2262 | 0.0136 | 0.0013 | 0.8644 | 0.0142 | 0.0014 | 3.4367 | 0.0144 | 0.0014 | 12.3295 | 0.0148 | 0.0014 | 91.4316 | 0.0312 | 0.0037 | 362.7264 | 0.1348 | 0.0152 |
| GC-RFM | 0.4968 | 0.3576 | 0.1545 | 8.5698 | 3.1702 | 0.3045 | 13.8962 | 6.2920 | 0.6044 | 28.3338 | 12.7096 | 1.2209 | 56.6676 | 25.4192 | 2.4418 | 114.3452 | 52.6177 | 4.9813 |
| GC-MEG | 0.6252 | 0.6252 | 0.2376 | 0.7253 | 0.7253 | 0.2758 | 1.3993 | 1.3993 | 0.5321 | 4.1523 | 4.1523 | 1.5788 | 16.3925 | 16.3925 | 6.2328 | 65.9471 | 65.9471 | 24.8062 |
| CF-PRE | 1.0232 | 1.0232 | 0.3802 | 2.5221 | 2.5221 | 0.9590 | 7.8329 | 7.8329 | 2.9783 | 22.8580 | 22.8580 | 8.6612 | 82.6153 | 82.6153 | 31.4127 | 327.9827 | 327.9827 | 124.3943 |
| CF-CAL | 3.8576 | 0.5796 | 0.0557 | 10.8208 | 1.3312 | 0.1279 | 12.9212 | 2.2774 | 0.2188 | 23.1592 | 4.5015 | 0.4324 | 45.6776 | 6.4552 | 0.6201 | 90.4416 | 16.2189 | 1.5657 |
| CF-SRT | 0.0332 | 0.0332 | 0.0126 | 0.0332 | 0.0332 | 0.0126 | 0.0332 | 0.0332 | 0.0126 | 0.0332 | 0.0332 | 0.0126 | 0.0332 | 0.0332 | 0.0126 | 0.0332 | 0.0332 | 0.0126 |
| CF-RES | 0.0028 | 0.0028 | 0.0011 | 0.0028 | 0.0028 | 0.0011 | 0.0028 | 0.0028 | 0.0011 | 0.0028 | 0.0028 | 0.0011 | 0.0028 | 0.0028 | 0.0011 | 0.0028 | 0.0028 | 0.0011 |

SAT: satellite; SAS: satellite server; GND: ground server. IS-LAD and -SEG indicate *imagery slicing*; CD-FMK and -GEN indicate *cloud detection*; GC-RFM and -MEG indicate *geometric correction*; CF-PRE, -CAL, -SRT, and -RES indicate *classification*.

TABLE VI

COMMUNICATION TIME COST (MS) INCURRED BY SEC TRANSMISSION LINKS

| Data transmission | 256×256 | | 512×512 | | 1024×1024 | | 2048×2048 | | 4096×4096 | | 8192×8192 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Inter-satellite | Satellite-to-ground | Inter-satellite | Satellite-to-ground | Inter-satellite | Satellite-to-ground | Inter-satellite | Satellite-to-ground | Inter-satellite | Satellite-to-ground | Inter-satellite | Satellite-to-ground |
| IS-LAD → IS-SEG | 0.0124 | 0.1081 | 0.0413 | 0.4434 | 0.0880 | 0.8832 | 0.1792 | 1.1336 | 0.5264 | 5.9046 | 0.9775 | 9.7475 |
| IS-SEG → CD-FMK | 0.0052 | 0.0558 | 0.0113 | 0.1350 | 0.0219 | 0.2331 | 0.0448 | 0.4625 | 0.0892 | 0.9046 | 0.1775 | 1.5941 |
| CD-FMK → CD-GEN | 0.0052 | 0.0558 | 0.0113 | 0.1350 | 0.0219 | 0.2331 | 0.0448 | 0.4625 | 0.0892 | 0.9046 | 0.1775 | 1.5941 |
| CD-GEN → GC-RFM | 0.0052 | 0.0558 | 0.0113 | 0.1350 | 0.0219 | 0.2331 | 0.0448 | 0.4625 | 0.0892 | 0.9046 | 0.1775 | 1.5941 |
| GC-RFM → GC-MEG | 0.0231 | 0.2176 | 0.0463 | 0.4722 | 0.0914 | 0.9269 | 0.1832 | 1.8665 | 0.3763 | 3.6470 | 0.7318 | 8.0510 |
| GC-MEG → CF-PRE | 0.0908 | 0.9232 | 0.3664 | 3.6720 | 1.4639 | 15.0022 | 5.8552 | 59.4185 | 23.4227 | 241.3250 | 93.6933 | 918.6355 |
| CF-PRE → CF-CAL | 0.0126 | 0.1431 | 0.0456 | 0.4722 | 0.0913 | 0.9270 | 0.3663 | 3.8018 | 0.7325 | 6.9377 | 1.4654 | 15.2371 |
| CF-CAL → CF-SRT | 0.0001 | 0.0012 | 0.0001 | 0.0012 | 0.0001 | 0.0012 | 0.0001 | 0.0012 | 0.0001 | 0.0012 | 0.0001 | 0.0012 |
| CF-SRT → CF-RES | 0.0001 | 0.0012 | 0.0001 | 0.0012 | 0.0001 | 0.0012 | 0.0001 | 0.0012 | 0.0001 | 0.0012 | 0.0001 | 0.0012 |

IS-LAD and -SEG indicate *imagery slicing*; CD-FMK and -GEN indicate *cloud detection*; GC-RFM and -MEG indicate *geometric correction*; CF-PRE, -CAL, -SRT, and -RES indicate *classification*.

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### C. Parameter Tuning

The performance of SECORS is influenced by two key parameters: the number of layers ($hp_l$) and the dimension of each layer ($hp_d$) in RNN, which are of pivotal importance in the scheduling decision-making process. The values of these two parameters are set to $hp_l = \{2, 4, 6\}$ and $hp_d = \{32, 64, 128\}$. Besides, different task scenarios are using three image sizes, i.e., 256 × 256, 1024 × 1024, and 4096 × 4096, and 5 satellites in SEC. Through extensive experiments in Table VII, we observe that a configuration of $hp_l = 4$ and $hp_d = 32$ demonstrates the optimal energy efficiency in the test scenario with a small image size of 256 × 256. For the scenario involving 4096 × 4096, the setting of $hp_l = 6$ and $hp_d = 32$ achieves the best result of time cost. Except for the above instances, the method configured with $hp_l = 4$ and $hp_d = 64$ performs a robust performance for SEC resource scheduling in a majority of instances, which notably improves the on-orbit imagery processing efficiency on time and energy. In the following experiments, the parameter setting $hp_l = 4, hp_d = 64$ is used to evaluate SECORS.

TABLE VII
RESULTS OF TIME (ms) AND ENERGY (Wms) COSTS ACHIEVED BY SEC-MPH WITH DIFFERENT PARAMETERS

| $hp_l$ | $hp_d$ | 256×256 (scenario 1) | | 1024×1024 (scenario 2) | | 4096×4096 (scenario 3) | |
|---|---|---|---|---|---|---|---|
| | | Time | Energy | Time | Energy | Time | Energy |
| 2 | 32 | 45.33 | 2217.09 | 567.69 | 2.90e+4 | 5046.12 | 4.48e+5 |
| | 64 | 39.75 | 2305.34 | 478.41 | 3.01e+4 | 4890.86 | 4.56e+5 |
| | 128 | 54.36 | 3389.24 | 604.23 | 3.89e+4 | 5679.55 | 5.46e+5 |
| 4 | 32 | 24.55 | 1089.43 | 286.14 | 1.52e+4 | 2237.27 | 1.79e+5 |
| | 64 | 12.31 | 1112.65 | 230.16 | 1.41e+4 | 1903.55 | 1.53e+5 |
| | 128 | 15.82 | 1273.06 | 236.10 | 2.08e+4 | 2450.89 | 2.76e+5 |
| 6 | 32 | 20.46 | 1652.39 | 267.01 | 2.25e+4 | 1884.23 | 1.67e+5 |
| | 64 | 17.60 | 2035.49 | 409.61 | 1.90e+4 | 2207.36 | 2.01e+5 |
| | 128 | 30.95 | 2330.57 | 503.18 | 2.95e+4 | 4330.52 | 3.28e+5 |

The best results of time and energy costs in each column are highlighted. $hp_l$ and $hp_d$ denote the number of layers and dimension of each layer, respectively in RNN.

### D. Performance of SEC-MPH

This section seeks to evaluate the efficacy of the proposed SEC-MPH for coping with the resource scheduling problem defined in this article. For all the instances, the online–offline condition in case 3 (Section III-A3) is chosen for SEC and the number of satellites is set to 5.

*1) Comparison to MADRL Algorithms:* We assess the comprehensive performance of our SEC-MPH and other three different MADRL-based approaches including COMA, VDAC, and LAIES, in this section. The total reward is chosen as the metric, and a larger reward represents better performance. As illustrated in Fig. 8, the convergence curve of each algorithm has been plotted, through which we can observe some key information. In cases of small and medium imagery processing scales, including 256, 512, and 1024, our method, VDAC, and LAIES, all show favorable reward and effective convergence. As the problem scale increases, the proposed SEC-MPH presents a remarkable superiority over others. In contrast, COMA is observed to exhibit bad capacity in solving the proposed SEC resource scheduling problem. Specifically, the fluctuation of the reward curve reflects the solution stability. From the stable perspective, our algorithm achieves the best performance from all scenarios, followed by VDAC, LAIES, and COMA. Furthermore, a faster convergence trait can be obtained by SEC-MPH in test instances involving 256, 1024, 2048, and 4096, where it begins to converge after approximately 2000 episodes. LAIES and VDAC demonstrate strong convergence for the test cases of 256 and 2048 but fail to ultimately converge to a satisfactory reward. The comparative results show that the proposed algorithms are able to make preferable SEC resource scheduling decisions for complex on-orbit imagery processing.

*2) Comparison to Multiobjective Evolutionary Algorithms:* The multiobjective problem-solving ability of the proposed algorithm SEC-MPH is evaluated through a comparative analysis with NSGA-II and MOEA/D. Since SEC-MPH obtains only one solution in a single run, we adjust the coefficient $\beta$ to generate multiple feasible solutions with different preference weights on time and energy cost, and the value range of $\beta$ is {0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9}. Subsequently, based on the nondominated extraction method [53], [54], the nondominated solutions are extracted from the above solutions. The results are visualized by plotting the Pareto front (PF) charts, as shown in Fig. 9, which illustrate two objectives (time and energy cost) of nondominated solutions in the same coordinate system. Our method achieves promisingly uniform and converged PF results when addressing all the SEC resource scheduling problems with varying data scales of on-orbit remote sensing imagery processing; NSGA-II shows a similar performance in distribution and convergence; followed by MOEA/D, which obtains inferior optimization results on time and energy costs. Although there is no obvious advantage revealed by our proposed algorithm in comparison to other classical multiobjective evolutionary algorithms, SEC-MPH is superior to NSGA-II and MOEA/D in computational efficiency, as demonstrated in the later context (Section V-E4). Therefore, the multiobjective solving capacity of SEC-MPH can be validated.

### E. Optimization Performance of SECORS on On-Orbit Remote Sensing Imagery Processing

In this section, we identify the optimization performance of the proposed method for on-orbit remote sensing imagery processing in the context of SEC. Evaluation metrics including time and energy cost, and energy efficiency ratio (EER) are adopted. EER is computed by dividing the number of pixels of an image by the consumed energy (Wms), which exposes the number of pixels that can be processed by the system per unit of energy and time cost. A larger EER value represents better performance. On top of that, in the cases of multiobjective algorithms (i.e., NSGA-II and MOEA/D) generating the set of solutions, a single-objective result for each solution is calculated using the weighted sum method with both the weights of two objectives set to 0.5. After the above transformation, the solution with the minimum objective value is taken as the final result.

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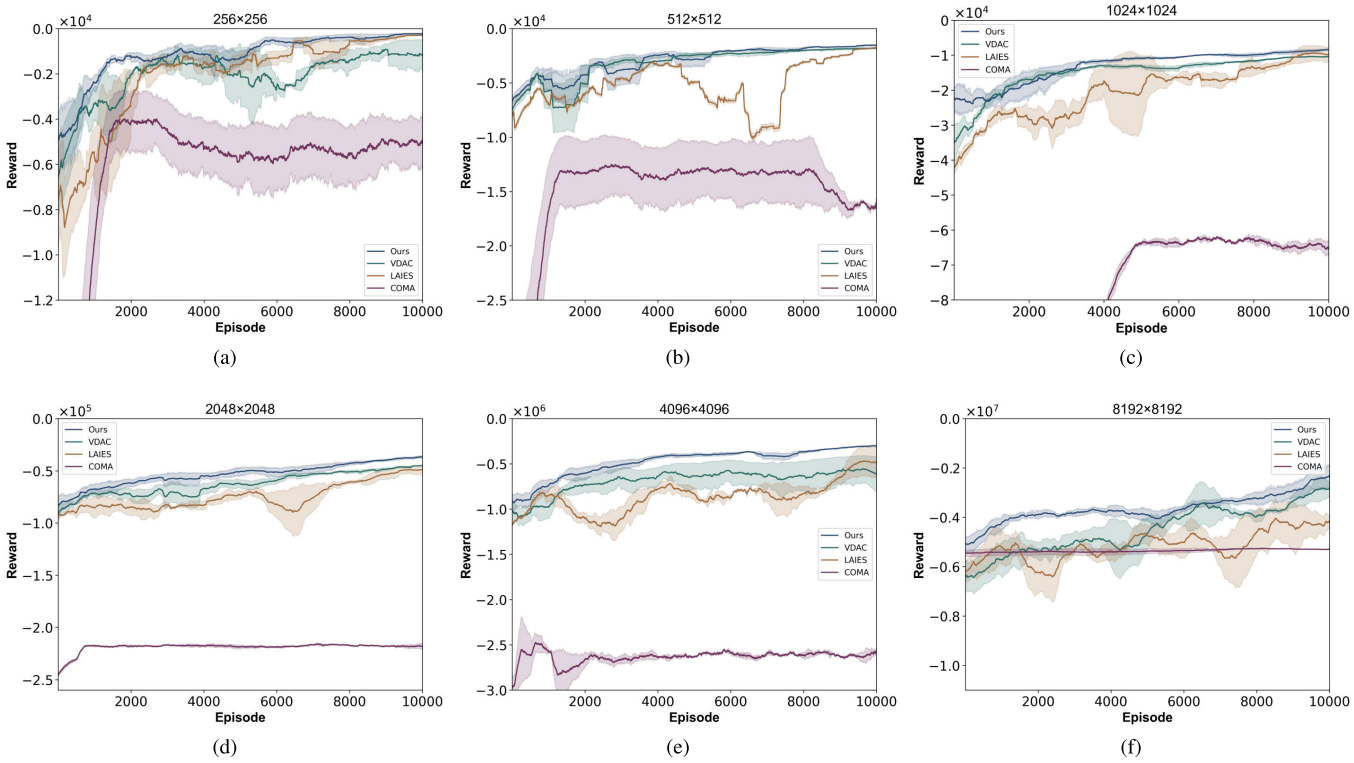



Fig. 8. Reward curves of each algorithm on SEC resource scheduling problems with five satellites and various image sizes. Each result is obtained by running the algorithm with five different random seeds; the darker line shows the mean over seeds; the shaded area indicates 80% confidence interval. (a)–(f) Reward results with imagery processing scales of 256 × 256, 512 × 512, 1024 × 1024, 2048 × 2048, 4096 × 4096, and8192 × 8192, respectively.

*1) Time and Energy Costs:* The experimental time and energy cost, as well as related savings (compared with baseline), are listed in Tables VIII and IX. In terms of time efficiency, as shown in Table VIII, our SECORS presents the superior performance for all the test cases; NSGA-II-based scheme acquires the second-best result, followed by MOEA/D, LAIES, and VDAC, and conversely, COMA shows an inferior optimizing performance. Particularly, the optimal performance of our method is observed in the 4096 × 4096 imagery processing scenario, with time reduction being up to nearly 69%. Next, as per the results obtained in Table IX, we can find that promising energy efficiency can be achieved by the proposed SEC-MPH in SECORS for most instances. NSGA-II, however, performs better than our method when optimizing the imagery processing with the image size set to 256 × 256. Subsequently, MOEA, LAIES, VDAC, and COMA follow the above two schemes. Note that our scheme continues to demonstrate the best optimization capacity in the 4096 × 4096 test case, bringing about an energy reduction of 66.60%. Overall, the proposed methodology is well-developed for SEC resource scheduling optimization, to achieve time- and energy-efficient on-orbit computing capacity.

*2) Energy Efficiency Ratio:* From Fig. 10, the following observations can be obtained. The satellite constellation based on our method can process more pixels per power and per time, outperforming other schemes in most instances. NSGA-II, similar to ours, manipulates a promising EER and executes better than others in the scenario of 256 × 256 imagery processing. Other schemes leveraging MOEA/D, LAIES, and VDAC also augment the on-orbit processing efficiency compared with the baseline (without any optimization). Suffering from the limitation of optimization presented above, COMA gains only a slight EER improvement over the baseline. Meanwhile, the EER results of all the methods rise as the image scale expands from 256 × 256 to 4096 × 4096. Particularly, our work endows satellites with more than 2.2× improvement in every case; it achieves the best EER of 548, 3.0× higher than the baseline. Therefore, the satellite constellation configured with the proposed methodology can keep high computational power and low energy cost, substantially enhancing the usability of complicated on-orbit remote sensing imagery processing.

*3) Communication Overhead:* Communication overhead is generated by data transmission during on-orbit remote sensing imagery processing through SEC, as shown in Fig. 11. Generally, for all the methods, the communication cost increases as the image size scales up and equates to about 10% of total time cost (demonstrated in Section V-E1). For the small-scale imagery processing lower than 2048 × 2048, the costs are within 200 ms, while the values of communication overhead exceed 1000 ms in the instance of 8192 × 8192. SEC-MPH generates the lowest cost, which benefits from its favorable solving performance, followed by NSGA-II, VDAC, and LAIES; COMA fails to obtain the promising communication efficiency. Meanwhile, our method controls the communication cost in a feasible range (≤200 ms) when handling the scheduling problems with image size

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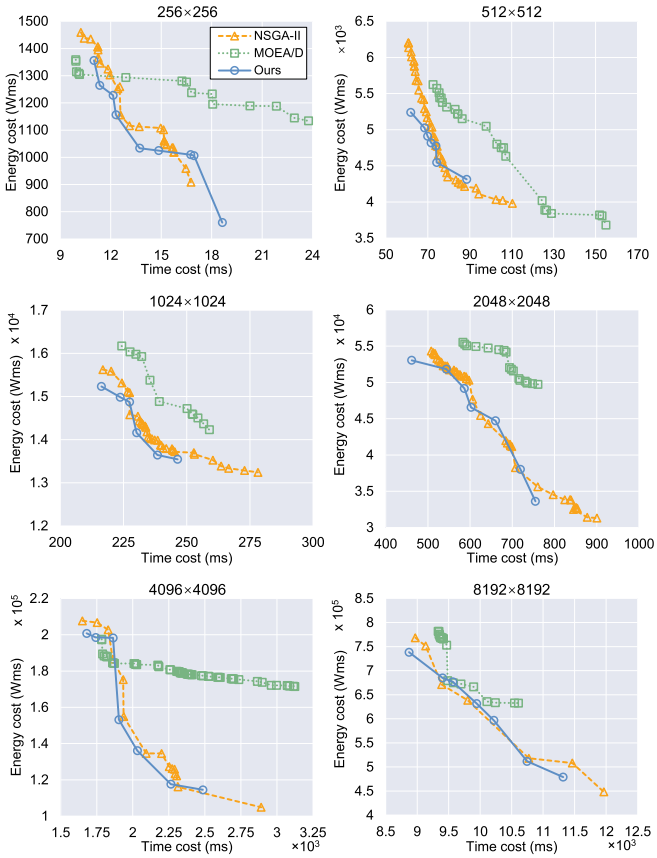



Fig. 9. PF results obtained by each algorithm on SEC resource scheduling problems with five satellites and various image sizes.

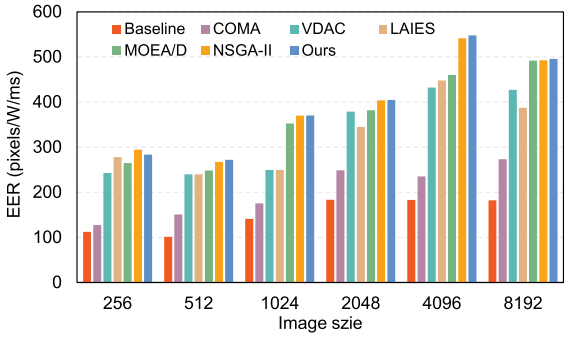



Fig. 10. EERs achieved by each method on on-orbit remote sensing imagery processing with five satellites and various image sizes.

smaller than 4096 × 4096. Thus, our method achieves a reasonable communication overhead on on-orbit imagery processing.

*4) Scheduling Overhead:* Scheduling overhead is another pivotal factor when evaluating the performance of optimization approaches. As illustrated in Fig. 12, EMO techniques exhibit a considerably higher overhead in comparison to MADRL approaches. This is due to the fact that MADRL leverages a training-then-solving paradigm, and then an end-to-end optimizer can be obtained for near-real-time execution. Specifically, MOEA/D and NSGA-II see a spike in the running overhead as the data scale increases, consuming up to 6000 ms and 4000 ms, respectively, in the 8192 × 8192 imagery processing instance. Conversely, our SEC-MPH, similar to other MADRL-based algorithms (i.e., COMA, VDAC, and LAIES), not only generates a lower running cost but also displays favorable stability within data scale variations. The overhead of the proposed SEC-MPH is maintained at a maximum of

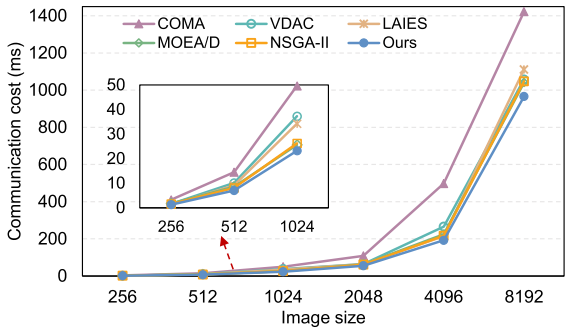



Fig. 11. Communication overhead incurred by each method.

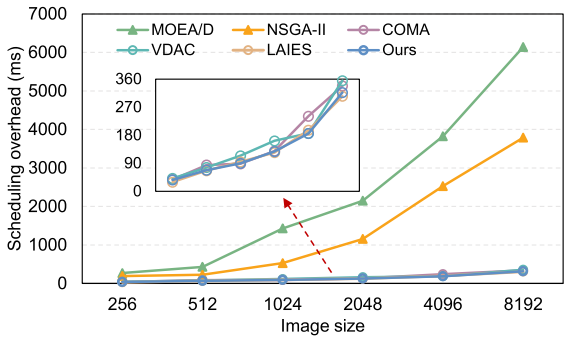



Fig. 12. Scheduling overhead incurred by each method.

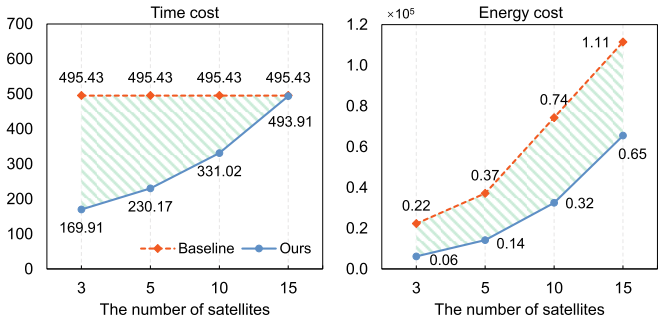



Fig. 13. Time and energy costs of on-orbit remote sensing imagery processing achieved by the proposed method with various satellite scales and image size set to 1024 × 1024.

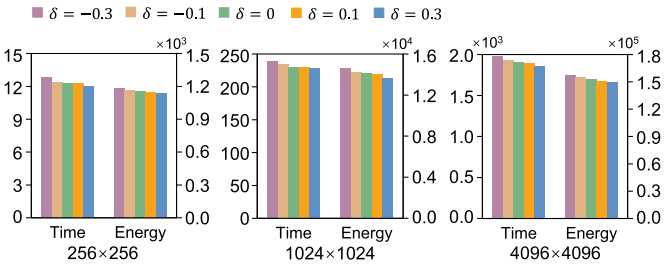



Fig. 14. Time and energy costs of on-orbit remote sensing imagery processing achieved by the proposed method with varying communication fluctuations.

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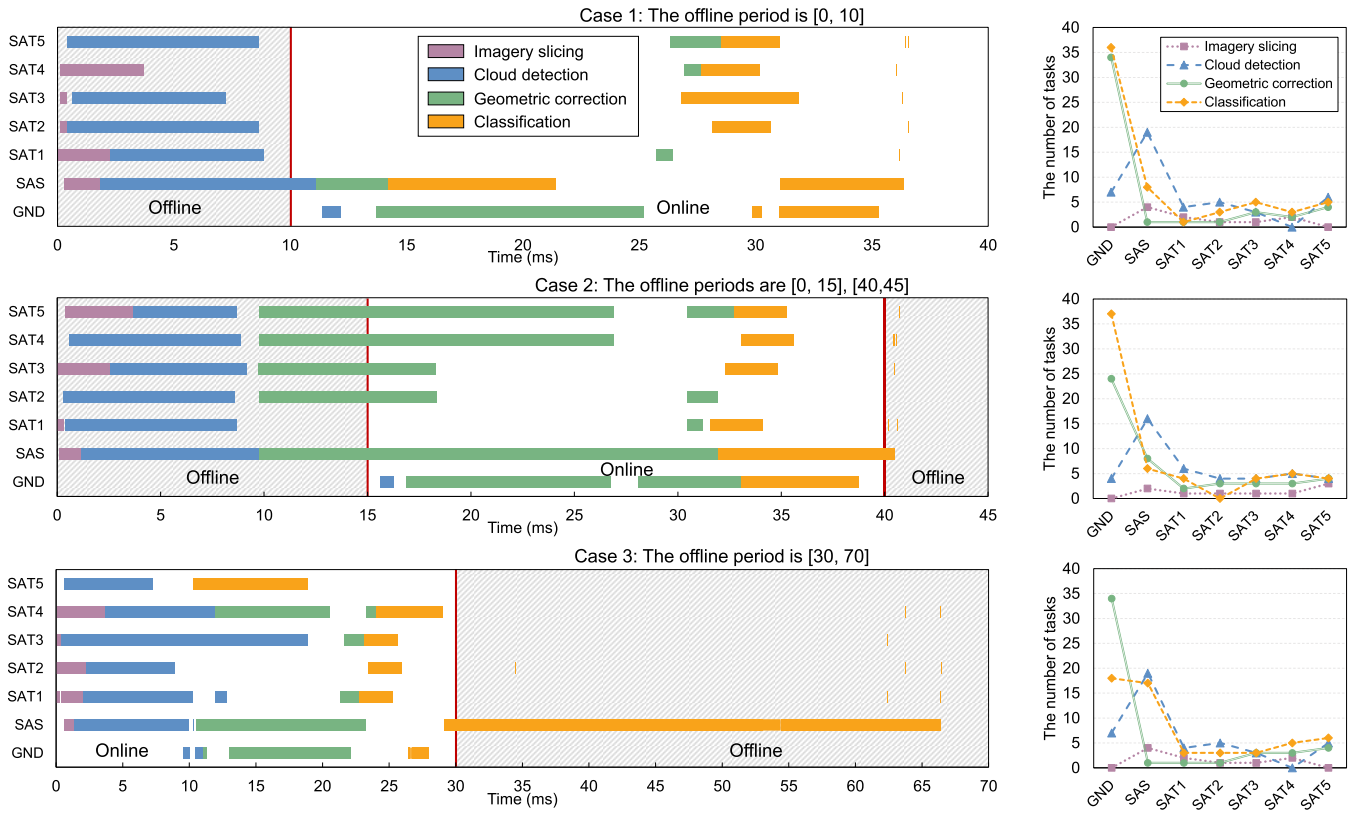



Fig. 15. Scheduling results on different online–offline scenarios. GND: ground server; SAS: satellite server; SAT: satellite.

360 ms for a range of on-orbit imagery processing cases (from 256 × 256 to 8192 × 8192). It can be concluded that the proposed method shows excellent scheduling efficiency.

### *F. Influence of Satellite Scale and Communication Interference*

This section first validates the on-orbit imagery processing performance of SECORS on varying satellite scales. Here, 1024 × 1024 is chosen as the image size. Fig. 13 illustrates the results of time and energy costs, where the green shaded area denotes the improvement of our method in comparison to the baseline. It can be observed that the time efficiency deteriorates as the number of satellites increases. Upon reaching 15 satellites, the time cost of the proposed method (493.91 ms) is nearly equivalent to that of the baseline (495.43 ms). The outcome can be attributed to the fact that one ground server and one satellite server are configured in SEC. As the satellite scale expands, the load of on-orbit image processing also rises, causing longer queuing time for tasks allocated to servers. Nevertheless, our method maintain a notable energy enhancement, achieving obvious energy reductions in all instances with satellite scale different in {3, 5, 10, 15}. It is explained as follows: although the increased on-orbit computing demands impose more waiting time for tasks to be executed by the server, the actual processing time on the server remains unchanged, and thus the energy cost is not influenced. From the above empirical evaluations with different satellite scales, this work reveals a promising SEC resource scheduling effectiveness for on-orbit imagery processing, particularly concerning the optimization of energy cost, which is crucial for such a power-constrained satellite computing environment.

Besides, we evaluate the optimization performance of SECORS under scenarios where intersatellite and satellite–ground communications are interfered with by the severe space environment. Here, referring to [55], we fluctuate the transmission rate using the following formulation:

$$B_{\delta}^{\text{rate}} = B^{\text{rate}} + \delta r B^{\text{rate}} \tag{31}$$

where $B^{\text{rate}}$, $\delta$, and $r$ represent the transmission rate, fluctuation coefficient, and random number uniformly distributed between $(0, 1)$, respectively. By setting $\delta = \{-0.3, -0.1, 0, 0.1, 0.3\}$ for transmission rates within intersatellite and satellite–ground simultaneously, we simulate real-world communication interference in SEC. The fluctuated transmission costs are then fed into SEC-RSM. Subsequently, we use the previously trained SEC-MPH to solve the problem, and the satellite scale and image size are set to 5 and {256 × 256, 1024 × 1024, 4096 × 4096}, respectively. Fig. 14 suggests that as the transmission rate varies, the optimization results of the proposed method exhibit slight differences. However, for each instance, the impact on time and energy costs caused by transmission fluctuations can be ignored due to the minimal variations. Therefore, the communication interference does not affect the scheduling decisions of the proposed method, whose robustness in optimizing on-orbit processing with dynamic and uncertain communication can also be validated.

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TABLE VIII
TIME COST (ms) AND SAVING (%) OF ON-ORBIT REMOTE SENSING IMAGERY PROCESSING WITH FIVE SATELLITES AND VARIOUS IMAGE SIZES

| Method | 256×256 | | 512×512 | | 1024×1024 | | 2048×2048 | | 4096×4096 | | 8192×8192 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Cost | Saving | Cost | Saving | Cost | Saving | Cost | Saving | Cost | Saving | Cost | Saving |
| Baseline | 38.92 | 0.00% | 172.76 | 0.00% | 495.43 | 0.00% | 1525.61 | 0.00% | 6113.88 | 0.00% | 2.45e+4 | 0.00% |
| COMA | 28.87 | 25.82% | 128.29 | 25.74% | 439.46 | 11.30% | 959.43 | 37.11% | 4403.91 | 27.97% | 1.70e+4 | 30.74% |
| VDAC | 14.04 | 63.94% | 91.14 | 47.24% | 335.77 | 32.23% | 575.23 | 62.30% | 2392.82 | 60.86% | 1.13e+4 | 53.87% |
| LAIES | 12.81 | 67.08% | 88.54 | 48.75% | 332.62 | 32.86% | 622.03 | 59.23% | 2081.52 | 65.95% | 1.27e+4 | 48.17% |
| MOEA/D | 16.80 | 56.83% | 83.01 | 51.95% | 239.16 | 51.73% | 613.91 | 59.76% | 2180.86 | 64.33% | 1.01e+4 | 58.86% |
| NSGA-II | 13.69 | 64.82% | 72.38 | 58.11% | 234.28 | 52.71% | 547.12 | 64.14% | 1936.32 | 68.33% | 9.80e+3 | 60.10% |
| Ours | **12.31** | **68.38%** | **67.60** | **60.87%** | **230.16** | **53.54%** | **543.69** | **64.36%** | **1903.55** | **68.87%** | **9.57e+3** | **61.06%** |

The best results of time cost and saving in each column are bold.

TABLE IX
ENERGY COST (Wms) AND SAVING (%) OF ON-ORBIT REMOTE SENSING IMAGERY PROCESSING WITH FIVE SATELLITES AND VARIOUS IMAGE SIZES

| Method | 256×256 | | 512×512 | | 1024×1024 | | 2048×2048 | | 4096×4096 | | 8192×8192 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Cost | Saving | Cost | Saving | Cost | Saving | Cost | Saving | Cost | Saving | Cost | Saving |
| Baseline | 2918.98 | 0.00% | 12957.25 | 0.00% | 3.72e+4 | 0.00% | 1.14e+5 | 0.00% | 4.59e+5 | 0.00% | 1.84e+6 | 0.00% |
| COMA | 2575.09 | 11.78% | 8701.21 | 32.85% | 2.99e+4 | 19.55% | 8.44e+4 | 26.27% | 3.57e+5 | 22.09% | 1.23e+6 | 33.34% |
| VDAC | 1351.44 | 53.70% | 5472.04 | 57.77% | 2.10e+4 | 43.40% | 5.54e+4 | 51.62% | 1.94e+5 | 57.65% | 7.86e+5 | 57.37% |
| LAIES | 1179.67 | 59.59% | 5465.43 | 57.82% | 2.10e+4 | 43.46% | 6.08e+4 | 46.84% | 1.87e+5 | 59.14% | 8.66e+5 | 52.99% |
| MOEA/D | 1237.39 | 57.61% | 5281.34 | 59.24% | 1.49e+4 | 59.94% | 5.50e+4 | 51.97% | 1.82e+5 | 60.22% | 6.83e+5 | 62.96% |
| NSGA-II | **1112.65** | **61.88%** | 4905.57 | 62.14% | 1.42e+4 | 61.83% | 5.19e+4 | 54.61% | 1.55e+5 | 66.21% | 6.81e+5 | 63.03% |
| Ours | 1156.05 | 60.40% | **4817.66** | **62.82%** | **1.41e+4** | **61.89%** | **5.18e+4** | **54.70%** | **1.53e+5** | **66.60%** | **6.77e+5** | **63.27%** |

### *G. Gantt Analysis on the Influence of the Online–Offline Mode*

This section demonstrates the impact of different online–offline scenarios on SECORS. For the test instance, we set the satellite scale and image size to 5 and 512 × 512, respectively. Gantt diagrams and task assignment charts are constructed to present scheduling results. As shown in Fig. 15, diverse solutions for scheduling SEC resources (including satellite itself, satellite server, and ground server) for on-orbit tasks from five satellites. Generally, in all cases, on-orbit tasks are executed locally or assigned to the satellite server when satellites are in offline status. Under normal communication conditions, the ground server is scheduled frequently to handle satellite tasks of *geometric correction* and *classification*, leading to less overhead. This is due to the high-performance GPU configured in the ground server, which can facilitate complex floating-point and matrix computations. Conversely, satellites and satellite server are used to execute tasks generated by *imagery slicing* and *cloud detection* missions.

In case 1, the system uses the satellite server to collaborate on-orbit processing during the beginning offline period. After 10 ms, it gradually begins to schedule ground resources to handle tasks from satellites, and 34 and 36 tasks of *geometric correction* and *classification* are mainly executed by the ground server, respectively. A certain number of *cloud detection* tasks are scheduled to the satellite server, while approximately five tasks from each on-orbit mission are executed by satellites themselves. The scheduling solution of case 2 is generally similar to case 1, but in this case, the ground server is scheduled to deal with substantially more *classification* tasks than *geometric correction* tasks. It can be observed from case 3 that when the satellite offline duration is extended, on-orbit tasks can only be executed locally or by the satellite server, increasing the total execution time.

Based on the above analysis, the proposed SECORS has the capacity to adjust to various offline periods, automatically allocating the appropriate device to tasks. Besides, our method can effectively make scheduling decisions to enhance the efficiency of on-orbit remote sensing imagery processing.

## VI. CONCLUSION AND FUTURE WORK

In this work, we have proposed an SEC resource scheduling method for efficient on-orbit remote sensing image processing, called SECORS. Specifically, we have first redesigned a remote-sensing-based SEC architecture considering offline–online operational mode. A resource scheduling model, SEC-RSM, involving task representation through DAG and mathematical formulation, has been established to describe the research problem. To tackle the problem and obtain effective scheduling solutions, we have developed an end-to-end algorithm, SEC-MPH, using MAPPO and scheduling heuristic of EFT. Then, we have also built a simulation SEC platform to conduct experiments. Empirical results reveal that SECORS can reduce execution time and energy costs by 68.87% and 66.60%, respectively, for on-orbit remote sensing imagery processing. In addition, our method achieves 3.0× enhancement in EER, endowing the satellite constellation with a favorable processing capability of 548 pixels per unit of power (W) and time (ms).

In the future, there are some substantial research issues worth exploring. With the rapid development of satellite constellations, we will investigate the multiple constellation collaboration for on-orbit computation. In light of the recent deployment of deep learning techniques on satellites, neural-network-based onboard processing will be considered for potential integration into our proposed scheduling framework. In addition, we plan to apply our method to the real-world satellite platform supported by China Satellite Network Group.

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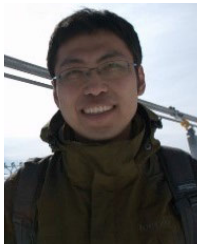
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